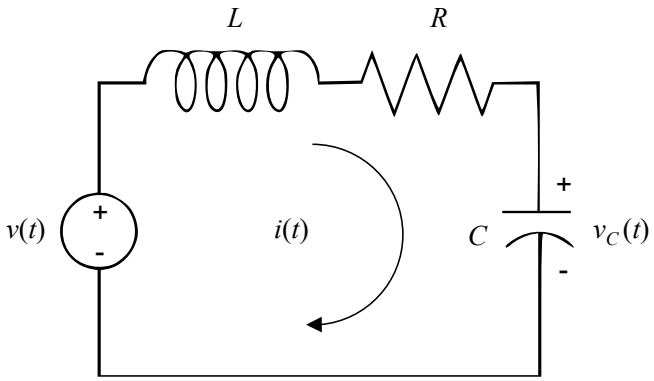
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Advanced Control System Design - Tutorial 2 – Solution

1. You have been asked previously in Tutorial 1 to find the transfer function that relates the capacitor voltage to the input voltage in the following circuit:



In this tutorial, you are required to do the following:

A. Find the state-space representation of the system. Use different state variables assumptions to get different state-space representations.

B. Find the state-space representation of the system when the values of R, L, and C are 1 Ω, 2 H, and 3 F, respectively.

C. Use MATLAB to compare the answer of this tutorial with the one you found in Tutorial 1.

D. Use Simulink to test the response of the system using any signal (a step signal, for instance).

Solution:

From Tutorial 1, we found that:

$$LC\frac{d^2v_C(t)}{dt^2}+RC\frac{dv_C(t)}{dt}+v_C(t)=v(t)$$

This can be rewritten as:

$$LC\ddot{v}_C+RC\dot{v}_C+v_C=v$$

Since $u=v,\ y=v_C$:

$$LC\ddot{y}+RC\dot{y}+y=u$$

Method 1:

$$x_1=v_C=y$$

$$x_2=\dot{v}_C=\dot{x}_1$$

$$\dot{x}_1=x_2 \tag{1}$$

$$\dot{x}_2=\ddot{v}_C=\ddot{y}=-\frac{1}{LC}x_1-\frac{R}{L}x_2+\frac{1}{LC}u \tag{2}$$

$$y=x_1 \tag{3}$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}0 & 1\\ -\frac{1}{LC} & -\frac{R}{L}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}0\\ \frac{1}{LC}\end{bmatrix}u=\begin{bmatrix}0 & 1\\ -0.17 & -0.5\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}0\\ 0.17\end{bmatrix}u$$

$$y=\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Method 2:

$$x_1=v_C=y$$

$$x_2=i$$

$$i=C\dot{v}_C$$

$$x_2=C\dot{v}_C$$

$$\dot{x}_1=\dot{y}=\dot{v}_C=\frac{1}{C}x_2 \tag{4}$$

$$\dot{x}_2 = C\ddot{v}_C = -\frac{1}{L}x_1 - \frac{R}{L}x_2 + \frac{1}{L}u \tag{5}$$

$$y = x_1 \tag{6}$$

$$\begin{bmatrix}\dot{x}_1 \\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & \frac{1}{C} \\ -\frac{1}{L} & -\frac{R}{L}\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix} + \begin{bmatrix}0 \\ \frac{1}{L}\end{bmatrix}u = \begin{bmatrix}0 & 0.33 \\ -0.5 & -0.5\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix} + \begin{bmatrix}0 \\ 0.5\end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix}$$

Transfer function:

$$\frac{V_C(s)}{V(s)} = \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{0.17}{s^2 + 0.5s + 0.17}$$

